SQL for Business Analysis

Why Do We Write SQL Queries?

Understanding the Business Meaning of Data

Prepared for DTank54 Group Students

Oracle HR Schema
Level: A1 (Beginner)

1. Why Do We Write SQL Queries?

SQL is a tool. But a tool for what? We write SQL queries to answer business questions. A company has a lot of data. This data is in tables. But data in tables is not useful by itself. We need to ask questions to get useful information from the data. This is called "data analysis".

Think about it like this: a doctor has many test results about a patient. The results are just numbers. But the doctor looks at these numbers and asks: "Is the patient healthy?" "Does the patient need medicine?" In the same way, a data analyst looks at company data and asks: "How much money do we pay our employees?" "Which department has the most people?" "Who are our highest paid workers?"

SQL is the language we use to ask these questions. Every SQL query starts with a business question. First, someone asks a question. Then, the data analyst writes a SQL query to find the answer. The answer helps the company make better decisions.

In this book, we will look at many business questions. For each question, we will see: what the business wants to know, why it is important, which tables we need, and what kind of SQL query can answer it. The queries go from easy to difficult.

2. What is Data Analysis?

Data analysis means looking at data to find useful information. When a manager says "I want to know our total salary cost this month", that is a business question. The data analyst writes a SQL query to find the answer. Without SQL, the analyst would need to look at every row in the EMPLOYEES table by hand. With a big company that has thousands of employees, this is impossible. SQL makes it fast and easy.

Here are some things data analysis can do for a company:

1. Find problems: For example, which employees earn less than the minimum salary for their job?

2. Make plans: For example, how many new people do we need to hire next year?

3. Save money: For example, which departments have the highest salary costs?

4. Help people: For example, which employees have not received a commission this year?

As you can see, SQL is not just about writing code. SQL is about helping a company understand its data and make good decisions. Every query you write has a purpose.

3. Types of Business Questions

Business questions usually fall into these categories. We will look at each one with examples from the Oracle HR schema.

4. Level 1 - Simple Reporting (Easy)

These are the most basic questions. A manager just wants to see some data. No filters, no conditions. Just "show me the data". These queries use only SELECT and FROM. They are the starting point of every data analysis.

4.1 "Show Me the List"

The simplest type of question. Someone wants to see a list of something. For example: "Give me a list of all our departments." Or: "Show me all job titles we have." These queries help managers get a quick overview of the company structure.

Business Question: "What departments do we have in our company?"

Why it matters: A new manager wants to know all departments. This helps them understand how the company is organized.

`Which tables: DEPARTMENTS`

SQL Skills Used

SELECT specific columns, FROM one table. This is the most basic SQL query. You choose which columns to show and from which table.

```
-- Show all departments
SELECT department_id, department_name
FROM departments;
```

Business Question: "What job titles do we have and what are the salary ranges?"

Why it matters: The HR team needs this information for job postings. They want to know: what salary can we offer for each job? This helps them write job ads and talk to candidates about pay.

`Which tables: JOBS`

```
-- Show all jobs with salary range
SELECT job_id, job_title,
min_salary, max_salary
FROM jobs;
```

Business Question: "Where are our office locations?"

Why it matters: A company is planning to open a new office. They need to see all current office locations. This helps them decide where to put the new office.

`Which tables: LOCATIONS`

```
-- Show all locations
SELECT location_id, city,
state_province, country_id
FROM locations;
```

4.2 "Who Works Here?"

These questions ask about people. A manager wants to know: who are our employees? What are their names? What do they do? This is basic employee reporting. Every HR department needs this type of information every day.

> **Business Question:** "Give me a list of all employees with their names and job IDs."

> **Why it matters:** This is the most common HR report. The HR team needs it for many things: checking employee records, sending emails, preparing reports for the boss.

`Which tables: EMPLOYEES`

```
-- Employee list with job info
SELECT employee_id, first_name, last_name,
email, job_id, salary
FROM employees;
```

> **Business Question:** "Who are the managers of each department?"

> **Why it matters:** An employee has a problem and needs to talk to their department manager. The HR team needs a quick list: which person manages which department? This is a very common request in any company.

`Which tables: DEPARTMENTS`

```
-- Departments and their managers
SELECT department_name, manager_id
FROM departments
WHERE manager_id IS NOT NULL;
```

5. Level 2 - Filtered Reports (Medium)

Now we add the WHERE clause. These questions are more specific. A manager does not want ALL data. They want only the data that matches a condition. For example: "Show me only the employees who earn more than 10000." This is where SQL becomes powerful. You can filter thousands of rows and see only what you need.

5.1 "Find People Who Match a Condition"

These are the most common business questions. A manager wants to find employees who match some rule. This is useful for salary reviews, performance checks, and many other HR tasks.

> **Business Question:** "Who are our highest paid employees?"

> **Why it matters:** The company wants to give a bonus to top earners. The finance team needs a list of employees who earn more than 12000 per month. This helps them plan the bonus budget.

`Which tables: EMPLOYEES`

SQL Skills Used

SELECT, FROM, WHERE with comparison operators (>, <, >=, <=, =, !=). You filter rows based on a condition.

```
-- Highest paid employees
SELECT first_name, last_name, salary
FROM employees
```

```
WHERE salary > 12000
ORDER BY salary DESC;
```

Business Question: "Which employees work in the Sales department (department 80)?"

Why it matters: The Sales director wants to know everyone on their team. They need to plan the work and set targets for each sales person. This is a simple filter by department.

Which tables: EMPLOYEES

```
-- Sales team members
SELECT first_name, last_name, salary
FROM employees
WHERE department_id = 80;
```

Business Question: "Which employees earn more than the maximum salary for their job?"

Why it matters: This is a very important question for HR and finance. Every job has a maximum salary. If someone earns more than the maximum, there might be a mistake. Or maybe this person got a special raise. The company needs to check this for salary audits.

Which tables: EMPLOYEES, JOBS (need JOIN - you will learn this later)

```
-- Note: This needs a JOIN (advanced topic)
-- You will learn JOINs in the next lessons
-- But the business question is clear:
-- "Is anyone being overpaid?"
```

5.2 "Who Does NOT Match?"

Sometimes the most important questions are about what is NOT there. "Who does NOT have a manager?" "Who does NOT get commission?" These questions help find problems in the data or missing information.

Business Question: "Which employees do NOT have a department?"

Why it matters: A manager is checking employee records. They find that some employees have no department assigned. This is a problem because every employee should belong to a department. The company needs to fix this data issue.

Which tables: EMPLOYEES

SQL Skills Used

WHERE with IS NULL. NULL means "no value" or "empty". This is different from zero. It means the data is missing.

```
-- Employees without department
SELECT first_name, last_name, department_id
FROM employees
WHERE department_id IS NULL;
```

Business Question: "Which departments do NOT have a manager assigned?"

Why it matters: The company is doing a review. They want to know: are there any departments without a manager? A department without a manager is a problem because nobody is leading that team. This query helps the company fix the issue.

`Which tables: DEPARTMENTS`

```
-- Departments without manager
SELECT department_name, manager_id
FROM departments
WHERE manager_id IS NULL;
```

Business Question: "Which employees earn a commission?"

Why it matters: The company wants to know who gets commission. Commission is extra money for sales people. The finance team needs this list to calculate total payments. If someone gets commission, the company pays them more than just their salary.

`Which tables: EMPLOYEES`

```
-- Employees with commission
SELECT first_name, last_name, salary,
commission_pct
FROM employees
WHERE commission_pct IS NOT NULL;
```

5.3 "Find by Pattern"

Sometimes you do not know the exact value. You only know part of it. For example: "I am looking for a customer whose name starts with S." This is when you use LIKE. It is very useful for searching in large tables.

Business Question: "Find all employees whose last name contains 'son' (like Johnson, Anderson)."

Why it matters: The HR team received a phone call from someone. The person said their last name has "son" in it, but the HR team forgot the full name. They need to search the employee database to find this person.

`Which tables: EMPLOYEES`

SQL Skills Used

WHERE with LIKE and wildcard symbols. % means any characters. _ means exactly one character.

```
-- Find last names with 'son'
SELECT first_name, last_name, email
FROM employees
WHERE last_name LIKE '%son%';
```

Business Question: "Find all departments whose name starts with 'Sa' (like Sales, Safety)."

Why it matters: A manager is writing a report and needs information about departments that start with "Sa". They are not sure if the department is called "Sales" or "Safety" or something else. They use LIKE to search.

Which tables: DEPARTMENTS

```
-- Departments starting with 'Sa'
SELECT department_name
FROM departments
WHERE department_name LIKE 'Sa%';
```

6. Level 3 - Analysis and Calculations (Harder)

Now we go beyond simple reporting. We do calculations. We combine data. We sort and organize results. These queries give the company real insights. They help managers make decisions.

6.1 "How Much Does It Cost?"

Money questions are very important. "How much do we pay in total?" "What is the annual salary cost?" These questions help the finance team plan budgets and control costs.

Business Question: "What is the total annual salary cost for each employee?"

Why it matters: The finance team needs to know the total cost of salaries for the whole year. Each employee has a monthly salary. The annual cost is monthly salary times 12. This helps the company plan the yearly budget. If the company has 100 employees and the average salary is 5000, the annual cost is about 6 million. This is a big number and the company needs to plan for it.

Which tables: EMPLOYEES

SQL Skills Used

Arithmetic in SELECT: *, +, -, /. Also column aliases. You can do math in SQL and give the result a name.

```
-- Annual salary for each employee
SELECT first_name, last_name,
salary AS "Monthly Salary",
salary * 12 AS "Annual Salary"
FROM employees
ORDER BY salary DESC;
```

Business Question: "What is the salary plus commission for each sales employee?"

Why it matters: The company wants to know the total cost of each sales person. Sales employees get a base salary plus commission. Commission is a percentage of sales. The total payment is: salary + (salary * commission_pct). This is important for budget planning because commission can change every month.

Which tables: EMPLOYEES

```
-- Total pay: salary + commission
SELECT first_name, last_name, salary,
commission_pct,
salary + salary * commission_pct AS "Total Pay"
FROM employees
WHERE commission_pct IS NOT NULL
```

```
ORDER BY "Total Pay" DESC;
```

6.2 "What Are the Ranges?"

These questions help managers understand the spread of data. "What is the lowest and highest salary?" "Which jobs pay between 5000 and 10000?" These questions help with salary planning and job market analysis.

> **Business Question:** "What are the minimum and maximum salaries across all jobs?"
>
> **Why it matters:** The HR team is updating the salary structure. They need to know: what is the lowest salary in the company? What is the highest? This helps them check if the salary ranges are fair. If the lowest salary is 2000 and the highest is 40000, the gap is very big. The company might need to adjust salaries.
>
> `Which tables: JOBS`

SQL Skills Used

WHERE with BETWEEN. BETWEEN includes both the start and end values.

```
-- Jobs with salary between 5000 and 15000
SELECT job_title, min_salary, max_salary
FROM jobs
WHERE min_salary BETWEEN 5000 AND 15000;
```

> **Business Question:** "Show me employees whose salary is NOT in the normal range (not between 5000 and 15000)."
>
> **Why it matters:** The company considers 5000 to 15000 as the normal salary range. They want to see employees who earn much less or much more than this range. Employees below 5000 might need a raise. Employees above 15000 are senior staff or managers.
>
> `Which tables: EMPLOYEES`

```
-- Salary outside normal range
SELECT first_name, last_name, salary
FROM employees
WHERE salary NOT BETWEEN 5000 AND 15000
ORDER BY salary;
```

6.3 "Ranking and Sorting"

Managers love rankings. "Who are the top 5 highest paid?" "Who was hired first?" "Who joined most recently?" These questions help with recognition, planning, and reporting.

> **Business Question:** "Show all employees sorted by salary from highest to lowest."
>
> **Why it matters:** The CEO wants to see who earns the most. They need a ranked list of all employees by salary. This helps them understand the salary structure and find any problems (like a junior employee who earns more than a senior employee).
>
> `Which tables: EMPLOYEES`

SQL Skills Used

ORDER BY ASC (ascending) and DESC (descending). You can sort by one or more columns.

```
-- All employees by salary (high to low)
SELECT first_name, last_name, job_id, salary
FROM employees
ORDER BY salary DESC;
```

Business Question: "Show employees sorted by department, then by salary within each department."

Why it matters: The HR team wants a report organized by department. Inside each department, they want to see employees sorted by salary. This way, they can easily compare salaries within the same department. It helps them check if salaries are fair within each team.

`Which tables: EMPLOYEES`

```
-- By department, then salary
SELECT department_id, first_name, last_name,
salary
FROM employees
ORDER BY department_id ASC, salary DESC;
```

6.4 "Show Me Unique Values"

Sometimes you do not need all the details. You just need a summary. "How many different departments do we have?" "What are all the different job IDs?" DISTINCT removes duplicates and gives you a clean list of unique values.

Business Question: "What are all the different job roles our employees have?"

Why it matters: The HR team wants to see all the different jobs in the company. They do not need to see every employee. They just need to see which jobs exist. For example: IT_PROG, SA_REP, FI_ACCOUNT, etc. This helps them understand what kind of workers the company has.

`Which tables: EMPLOYEES`

SQL Skills Used

DISTINCT removes duplicate rows. You can use it with one or more columns.

```
-- Unique job IDs in the company
SELECT DISTINCT job_id
FROM employees;
```

Business Question: "What are the different city-country combinations where we have offices?"

Why it matters: The company is planning travel for a conference. They need to know: in which cities and countries do we have offices? They need a unique list, not the same city repeated many times.

`Which tables: LOCATIONS`

```
-- Unique city-country pairs
SELECT DISTINCT city, country_id
FROM locations
ORDER BY city;
```

7. Level 4 - Advanced Business Questions (Challenging)

These questions combine many SQL skills. They are closer to real-world business problems. You need to think carefully about which tables, which columns, and which SQL features to use.

> **Business Question:** "Which employees in the Sales department earn more than 10000 and also get a commission?"

Why it matters: The Sales director is planning the team structure. They want to know: which sales people are top performers? Top performers earn more than 10000 AND they get commission. These are the people who make the most sales. The company wants to give them special training or rewards.

`Which tables: EMPLOYEES`

SQL Skills Used

WHERE with AND, comparison operators, IS NOT NULL. Multiple conditions combined together.

```
-- Top sales performers
SELECT first_name, last_name, salary,
commission_pct
FROM employees
WHERE department_id = 80
AND salary > 10000
AND commission_pct IS NOT NULL
ORDER BY salary DESC;
```

> **Business Question:** "Find employees whose first name starts with 'S' or 'A' and who earn between 5000 and 15000."

Why it matters: The HR team is sending invitations to a company event. They want to invite employees whose names start with S or A (for a group activity) and who earn between 5000 and 15000. This is a specific filter that combines pattern matching with range checking.

`Which tables: EMPLOYEES`

SQL Skills Used

WHERE with LIKE, OR, AND, BETWEEN. Combining pattern matching with ranges.

```
-- Complex filter
SELECT first_name, last_name, salary
FROM employees
```

```
WHERE (first_name LIKE 'S%' OR first_name LIKE 'A%')
AND salary BETWEEN 5000 AND 15000
ORDER BY first_name;
```

Business Question: "Create an employee directory: show full name, job title info, and formatted output."

Why it matters: The company wants to print an employee directory. It should show the full name (first + last name joined together), the email, and the salary formatted as "Monthly: XXXX, Annual: XXXXX". This is a presentation task. The data needs to look nice and professional.

`Which tables: EMPLOYEES`

SQL Skills Used

String concatenation with ||, arithmetic, aliases with double quotes.

```
-- Employee directory
SELECT first_name || ' ' || last_name AS "Full Name",
email,
salary AS "Monthly",
salary * 12 AS "Annual"
FROM employees
ORDER BY "Full Name";
```

Business Question: "Show the job history of employees - which employees changed their jobs?"

Why it matters: The HR team wants to understand employee career paths. Which employees changed their jobs? When did they change? From which job to which job? This information helps with career planning and understanding employee growth.

`Which tables: JOB_HISTORY, EMPLOYEES, JOBS`

```
-- Note: Full query needs JOINs (next lessons)
-- But the business question is:
-- "How do employees grow in our company?"

SELECT employee_id, start_date, end_date,
job_id, department_id
FROM job_history
ORDER BY employee_id, start_date;
```

8. Complete List: SQL Queries by Business Purpose

This table shows all the business questions we discussed, organized by difficulty level. Use this as a reference when you practice SQL.

#	Level	Business Question	Key SQL Skill
1	Easy	What departments do we have?	SELECT, FROM

2	Easy	What are the job titles and salary ranges?	SELECT specific columns
3	Easy	Where are our offices?	SELECT, FROM
4	Easy	Who works here? (employee list)	SELECT, FROM
5	Easy	Who are the department managers?	WHERE IS NOT NULL
6	Medium	Who are the highest paid employees?	WHERE >, ORDER BY
7	Medium	Who works in the Sales department?	WHERE =
8	Medium	Who earns more than the max salary?	WHERE > (needs JOIN)
9	Medium	Who has no department?	WHERE IS NULL
10	Medium	Which department has no manager?	WHERE IS NULL
11	Medium	Who earns a commission?	WHERE IS NOT NULL
12	Medium	Find employee by name pattern	WHERE LIKE
13	Medium	Find department by name pattern	WHERE LIKE
14	Hard	What is the annual salary cost?	Arithmetic (*, /)
15	Hard	Total pay including commission?	Arithmetic + IS NOT NULL
16	Hard	Jobs in a salary range	WHERE BETWEEN
17	Hard	Salaries outside normal range	WHERE NOT BETWEEN
18	Hard	Rank employees by salary	ORDER BY DESC
19	Hard	Salary by department ranking	ORDER BY multi-column
20	Hard	What job roles exist in the company?	DISTINCT
21	Hard	Unique office locations	DISTINCT multiple
22	Challenging	Top sales performers (>10000 + commission)	AND, IS NOT NULL
23	Challenging	Name pattern + salary range filter	LIKE + BETWEEN + OR
24	Challenging	Create employee directory	||, arithmetic, alias
25	Challenging	Employee job change history	JOB_HISTORY table

Table 1: Complete SQL Business Queries List

9. Key Lesson

Remember this: every SQL query starts with a business question. Nobody writes SQL just for fun. Someone in the company needs information to make a decision. Your job as a data analyst is to understand the business question and write the right SQL query to answer it.

Before you write any SQL query, always ask yourself these three questions:

1. What does the business want to know? (The business question)

2. Which tables have the data I need? (The data source)

3. What conditions or calculations do I need? (The SQL logic)

If you can answer these three questions, you can write the SQL query. Practice these questions with the HR schema. The more you practice, the better you will understand both SQL and business analysis. Good luck, DTank54!